

Technology Stack

Date	5-7-2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies used to develop the EduGenie AI Learning Assistant, including the programming languages, frameworks, frontend and backend tools, database, AI APIs, and supporting software. A suitable technology stack ensures that the application is efficient, maintainable, scalable, and capable of delivering AI-powered educational features such as question answering, concept explanation, quiz generation, text summarization, and personalized learning recommendations. Fill out the table below with the technologies used in each architectural component and provide a brief justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2	Builds a responsive and user-friendly interface and dynamically displays AI-generated results.

2	Backend / Server-Side	<i>Python, FastAPI</i>	Processes user requests, integrates AI modules, and provides fast API responses.
3	Database / Data Storage	<i>JSON Files</i>	Stores user queries, quizzes, summaries, and learning data in a lightweight format.
4	Cloud / Hosting / Deployment	<i>Uvicorn (Local Deploymen)</i>	Manages source code, version history, and enables collaboration among team members.
5	Version Control & CI/CD	<i>GitHub</i>	Used for version control, code management, and collaboration among team members.

6	Third-Party APIs / Other Tools	<i>Google Gemini 2.5 Flash API</i>	Generates AI-powered answers, explanations, quizzes, summaries, and personalized learning recommendations.